



R V COLLEGE OF ENGINEERING
(An autonomous institution affiliated to VTU, Belgaum)
DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

IM233AI – WORK SYSTEMS DESIGN

UNIT V

Topic ID- 13 : Introduction to Ergonomics & Human Factors

Q No	Question	Marks	Difficulty Level	BT Level	COs
1.	What is meant by the term Ergonomics and discuss its primary objectives.	8	L	2	4
2.	Describe the components of a Human-Machine System.	8	M	2	4
3.	Discuss the "Human as a System Component" including strengths and limitations.	8	M	4	4
4.	Explain the three major domains (themes) of Ergonomics. Discuss how each domain contributes to the optimization of human well-being and overall system performance.	8	M	4	4
5.	Explain the role of Anthropometry in work system design.	8	H	4	4
6.	Describe the Human Information Processing Model.	8	M	2	4
7.	Analyze the impact of Illumination on work performance.	8	M	4	4
8.	Discuss Noise and its control in the workplace.	8	M	4	4
9.	Discuss "Environmental Stressors" and their cumulative effect on productivity.	8	M	4	4
10.	Discuss the "Fitting the Task to the Man" (FTTM) philosophy. Why is this approach superior to "Fitting the Man to the Task" (FMTT)?	8	M	3	4

Topic ID-14: Occupational Safety and Health

11.	How can "Poka-Yoke" (Error Proofing) be used as an ergonomic strategy to prevent industrial accidents? Give two examples.	8	M	3	4
12.	Differentiate between "Unsafe Acts" and "Unsafe Conditions." Which is generally more prevalent in industrial accidents, and why?	8	M	3	4



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13.	What are Work-related Musculoskeletal Disorders (WMSDs)? List the primary risk factors.	8	M	2	4
14.	Describe the "Reach Envelope" and its importance in workstation layout.	8	M	3	4
15.	Define "Industrial Accident" and differentiate between "Accident" and "Injury"	8	M	2	4
16.	Analyze the role of "Unsafe Acts" as a primary cause of accidents in workplaces.	8	M	4	4
17.	Discuss the impact of "Human Factors" on accident rates.	8	M	3	4
18.	What is meant by "Occupational Disease" and how it differs from an "Industrial Injury."	8	M	2	4
19.	Develop an "Ergonomic Checklist" for an office worker to self-assess their risk for developing Repetitive Strain Injuries (RSI).	8	H	5	4
20.	What are "Near Misses" and why are they critical for industrial safety management?	8	M	2	4
21.	Describe "Unsafe Conditions" in the context of workplaces and provide four workplace examples.	8	M	3	4